

4. The UK relies on a continuous supply of electricity which is provided by a variety of different power stations. Electricity can be generated in power stations and transmitted to users through the National Grid.

One concern with electricity production is the amount of greenhouse gases produced. Increased levels of these gases in our atmosphere is leading to climate change. Renewable energy resources produce less greenhouse gases but are not able to provide a constant supply of electricity.

(a) (i) **Draw a line** from each method of electricity generation on the left to the most likely disadvantage on the right. [2]

Method of generating electricity

Nuclear power station
Wind farm
Coal-fired power station

Disadvantage

High CO ₂ emissions
High decommissioning cost
Unreliable

(ii) The energy input to a nuclear power station is 20 MJ and its efficiency is 40%. The energy input to a coal-fired power station is 20 MJ and its useful output energy is 8 MJ.

A student suggests that the nuclear power station is more efficient than the coal-fired power station as it wastes less energy. Explain whether or not you agree with this conclusion. (You may wish to use an equation from page 2.) [2]

Space for calculations

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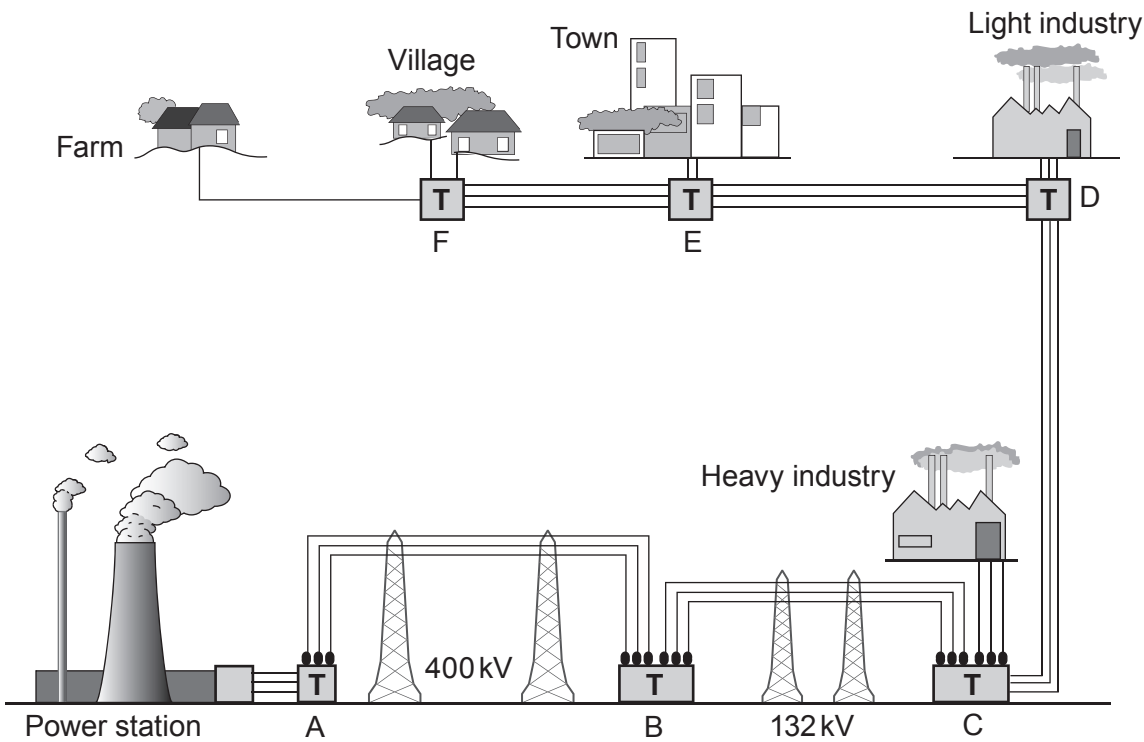
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(b) The diagram shows a part of the National Grid.

Key

T – transformer



(i) State **one** advantage of the National Grid. [1]

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(ii) Which of the transformers A, B, C, D, E or F is a step-up transformer? [1]

(iii) Explain why step-up transformers are used in the National Grid. [2]

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